Total Urinary Porphyrin (TUP)

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Identity Sheet

Property	Value
Name	Total Urinary Porphyrins (TUP)
Other names	Urine porphyrin fractionation, urinary porphyrin panel
Chemical formula	Variable — porphyrins are macrocyclic tetrapyrroles; core porphine ring: $C_{20}H_{14}N_4$
Molecular weight	Uroporphyrin: 830.76; Coproporphyrin: 654.71; Protoporphyrin: 562.66 g/mol
CAS numbers	Uroporphyrin I: 607-14-7; Coproporphyrin I: 531-15-7; Coproporphyrin III: 14643-77-7
Appearance	Intensely coloured (red-purple) fluorescent compounds

Structural formula (porphine core):

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N N
/  /  H-C C—C C-H
| || | | || |
H-C C C C-H
 / |  /
N—Fe—N (metalloporphyrin)
/  | /  H-C C C C-H
| || | | || |
H-C C—C C-H
 /  /
N N
```

TUP is a composite measure of all porphyrins excreted in urine, primarily **uroporphyrin** (octacarboxylic, 8-COOH), **heptacarboxylporphyrin** (7-COOH), **hexacarboxylporphyrin** (6-COOH), **pentacarboxylporphyrin** (5-COOH), and **coproporphyrin** (tetracarboxylic, 4-COOH). These are intermediates or by-products of the heme biosynthetic pathway. All porphyrins share the characteristic **Soret absorption band** (~400 nm) and red fluorescence (emission ~595–630 nm), making them uniquely suited for spectroscopic detection. They are anionic at urinary pH due to multiple carboxyl groups.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Porphobilinogen (PBG)	$C_{10}H_{14}N_2O_4$	226.23	Monopyrrole precursor, not a porphyrin; no Soret band or fluorescence

Molecule	Formula	MW (g/mol)	Key Difference
Bilirubin	$C_{33}H_{36}N_4O_6$	584.66	Linear tetrapyrrole (open ring), not cyclic; yellow not red; different metabolism
Protoporphyrin IX	$C_{34}H_{34}N_4O_4$	562.66	Predominantly in erythrocytes/faeces, minimal urinary excretion (too hydrophobic)
Heme (iron protoporphyrin IX)	$C_{34}H_{32}FeN_4O_4$	616.49	Metal-complexed porphyrin; fluorescence quenched by iron

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Medical Information

Origin

Endogenous: Biosynthesis — Heme Pathway Porphyrins are synthesised in all nucleated cells as part of the **heme biosynthesis pathway** (8 enzymatic steps from glycine + succinyl-CoA to heme). The major urinary porphyrins (uroporphyrin and coproporphyrin) arise from:

1. **ALA synthase** produces δ-aminolevulinic acid (ALA) — rate-limiting step.
2. **ALA dehydratase** condenses 2 ALA → porphobilinogen (PBG).
3. **PBG deaminase** polymerises 4 PBG → hydroxymethylbilane.
4. **Uroporphyrinogen III synthase** cyclises → uroporphyrinogen III (type III isomer, physiological).
5. **Uroporphyrinogen decarboxylase** → coproporphyrinogen III.
6. Subsequent steps yield protoporphyrinogen IX → protoporphyrin IX → heme (with iron insertion by ferrochelatase).

Urinary porphyrins are oxidised by-products: uroporphyrinogen and coproporphyrinogen spontaneously oxidise to uroporphyrin and coproporphyrin (which cannot be utilised for heme synthesis). Both type I (non-physiological) and type III isomers are excreted. Major production sites: **bone marrow** (erythropoiesis) and **liver**.

Exogenous Porphyrins are not absorbed from diet in significant amounts. Dietary chlorophyll (a magnesium porphyrin) is structurally related but metabolised differently and does not contribute to urinary porphyrin excretion.

Primary & Secondary Biological Roles

Primary role: - **Heme precursors:** Porphyrinogens are essential intermediates in heme biosynthesis, required for haemoglobin, cytochromes, and other haemoproteins.

Secondary roles: - **No independent physiological role.** Free porphyrins are waste products from non-enzymatic oxidation of porphyrinogens. They are phototoxic — upon absorption of light (Soret band ~400 nm), they generate reactive oxygen species, causing tissue damage in cutaneous porphyrias.

Catabolism and Elimination Pathway

- **Uroporphyrin** (8 carboxyl groups, highly hydrophilic): excreted **predominantly in urine**.
- **Coproporphyrin** (4 carboxyl groups, intermediate hydrophilicity): excreted in both **urine and bile/faeces** (~60% urine, ~40% faeces).

- **Protoporphyrin** (2 carboxyl groups, hydrophobic): excreted almost exclusively in **faeces via bile**; virtually absent from urine.
- Porphyrins are freely filtered at the glomerulus; coproporphyrin undergoes some tubular secretion.
- There is no significant hepatic catabolism of porphyrins; they are excreted intact.

Expression in Humans

Normal Levels

Compartment	Reference Range
Total urinary porphyrins (24-h)	< 300 µg/day (some labs: < 200 µg/day)
Uroporphyrin, urine	< 30 µg/day (< 36 nmol/day)
Coproporphyrin, urine	< 200 µg/day (< 300 nmol/day)
Heptacarboxylporphyrin, urine	< 10 µg/day
Hexa- and pentacarboxylporphyrins	< 5 µg/day each
Coproporphyrin III/I ratio	Normally III > I (ratio ~2.5–4:1)

Factors Influencing Levels **Increased TUP:** - Porphyria cutanea tarda (PCT) — marked uroporphyrin elevation - Acute intermittent porphyria (AIP) — moderate coproporphyrin increase during attacks - Variegate porphyria (VP) — coproporphyrin elevation - Hereditary coproporphyria (HCP) — coproporphyrin III markedly elevated - Lead poisoning (coproporphyrin elevated, ALA elevated) - Liver disease (alcoholic, hepatitis) — secondary coproporphyrinuria - Heavy metal exposure (arsenic, mercury)

Decreased TUP: - Not clinically significant - Low protein intake (reduced heme turnover) - Iron deficiency (reduced heme synthesis) - Adequate nutritional status - No known pathology associated with low urinary porphyrins

Associated Pathologies

Condition	TUP Pattern	Key Symptoms
Porphyria cutanea tarda (PCT)	Uroporphyrin »coproporphyrin; TUP >1000 µg/day	Photosensitive skin blistering, fragility, hypertrichosis; prevalence ~1:10,000
Acute intermittent porphyria (AIP)	Moderate coproporphyrin rise during attacks; PBG markedly elevated	Abdominal pain, neuropsychiatric symptoms; ~1:20,000
Hereditary coproporphyria (HCP)	Coproporphyrin III markedly elevated	Acute attacks + occasional photosensitivity; ~1:100,000
Variegate porphyria (VP)	Coproporphyrin > uroporphyrin	Acute attacks + skin photosensitivity; ~1:100,000 (except South Africa ~1:300)
Congenital erythropoietic porphyria (CEP)	Massive uroporphyrin I elevation	Severe photosensitivity from infancy, red urine; extremely rare
Lead poisoning	Coproporphyrin elevated (predominantly type III)	Abdominal colic, anaemia, neuropathy

Presence in Urine

Should they be normally present? Yes, in **small amounts**. Normal urinary excretion of total porphyrins is < 200–300 µg/day. Coproporphyrin is the dominant urinary porphyrin in health.

Normal urinary levels: Total < 300 µg/day; coproporphyrin < 200 µg/day; uroporphyrin < 30 µg/day.

Form in urine: Porphyrins are excreted as free (non-metal-complexed), oxidised porphyrins in their **anionic forms** (deprotonated carboxyl groups at urinary pH). They are intensely coloured (red-purple) and fluorescent.

Pathological significance:

Urinary Porphyrins	Possible Causes	Prevalence
Elevated (total >300 µg/day)	Porphyrias (PCT, AIP, VP, HCP, CEP), lead poisoning, liver disease, heavy metals	PCT: ~1:10,000; lead exposure varies
Decreased	No clinical significance	—

Solubility: Uroporphyrin is highly water-soluble (8 carboxyl groups). Coproporphyrin is moderately soluble. Protoporphyrin is poorly water-soluble (excreted in faeces, not urine). Porphyrins do not form urinary crystals.

Detection in Urine

Available Clinical Assays

1. **Spectrophotometric scanning (Soret band):**
   - **Principle:** Acidified urine extract scanned by UV-Vis spectrophotometer; total porphyrins quantified by Soret band absorption.
   - **Detection:** Absorption peak at **~400–405 nm** (Soret band) in acidic solution (0.5 N HCl).
   - **LOD:** ~10–20 µg/L.
   - **Advantages:** Simple, rapid screening.
   - **Disadvantages:** Does not differentiate individual porphyrin species.
2. **HPLC with fluorescence detection (gold standard for fractionation):**
   - **Principle:** Reversed-phase HPLC separates individual porphyrins (uro-, hepta-, hexa-, penta-, copro-) by carboxyl number.
   - **Detection:** Fluorescence — **Ex 400–410 nm / Em 618–625 nm**.
   - **LOD:** ~1–5 µg/L per species.
   - **Advantages:** Full fractionation, quantitative, identifies isomers (I vs III).
   - **Disadvantages:** Instrumentation cost, ~30 min run time.
3. **Qualitative fluorescence screening (Wood's lamp):**
   - **Principle:** Acidified urine examined under long-wave UV light (Wood's lamp, 365 nm); porphyrins fluoresce pink-red.
   - **Detection:** Visual fluorescence at **Em ~595–630 nm** under **Ex 365 nm**.
   - **LOD:** ~100–200 µg/L (insensitive).
   - **Advantages:** Instant bedside screening.
   - **Disadvantages:** Qualitative only; many interfering fluorescent substances.

Optimal Urine Type for Measurement

- **24-hour urine** is preferred for quantitative porphyrin measurement (results expressed as µg/day).
- Random (spot) urine can be used with creatinine correction.
- **Protection from light is essential** — porphyrins are photodegradable. Collect in dark/opaque containers.
- Preserve at **pH <7** (add 5 g sodium carbonate per 24-h container, OR keep at pH 6–7 with small amount of sodium bicarbonate — lab-dependent); store at **2–8 °C**.
- First morning urine offers higher concentration but no diagnostic advantage over 24-h.

Actual Gold Standard

HPLC with fluorescence detection (Ex 400 nm / Em 620 nm) on reversed-phase C18 columns is the reference method for urinary porphyrin fractionation. Used on Shimadzu, Agilent, and Waters platforms. Provides individual quantification of uroporphyrin, heptacarboxyl-, hexacarboxyl-, pentacarboxyl-, and coproporphyrin with isomer resolution (I vs III). CV: 5–10%. Reference labs: Mayo Clinic, ARUP, Quest.

Interferences in Measurement

Interference	Effect	Mechanism
Light exposure	False decrease	Photodegradation of porphyrins
Alkaline pH	False decrease	Accelerated degradation
Antibiotics (fluoroquinolones)	False positive (fluorescence)	Intrinsic fluorescence interferes
Rifampicin	False positive (colorimetric)	Orange colour overlaps with porphyrin absorbance
Bilirubin (high levels)	Spectral interference	Absorption near 450 nm may affect baseline
Bacterial contamination	False decrease	Bacterial metabolism of porphyrins
Dietary beetroot / food dyes	False positive (visual screening)	Red pigments mimic porphyrin fluorescence

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis (Soret band):** The defining feature of porphyrins. All porphyrins show an intense Soret (B) band at **~398–405 nm** ($\varepsilon \approx 3–5 \times 10^5$ $M^{-1}cm^{-1}$) and weaker Q bands at **~500, 530, 565, 620 nm**.
  - Dual-wavelength spectrophotometry (Soret peak vs baseline at 430 nm) for TUP quantification.
  - LOD: ~5–10 µg/L in acidified urine extracts.
  - Sample prep: Acidification to pH 1–2 (HCl), extraction with ethyl acetate/acetic acid, back-extraction into HCl.
- **NIR:** Not useful — porphyrins have no significant NIR absorption at clinically relevant concentrations.

Fluorescence Detection Porphyrins are **intensely fluorescent**, making this the most sensitive detection modality.

- **Direct fluorescence spectroscopy:** Ex 398–410 nm / Em 595–630 nm. The Soret band excitation yields characteristic two-peak red emission (uroporphyrin: Em 596 + 653 nm; coproporphyrin: Em 618 + 676 nm).
  - LOD: ~0.1–1 µg/L (sub-nanomolar sensitivity).
  - Sample prep: Acidification; optionally extraction to remove matrix quenchers.
- **HPLC-fluorescence:** Ex 400 nm / Em 618 nm — gold standard fractionation.
  - LOD: ~1 µg/L per species.
- **Time-resolved fluorescence:** Porphyrins have long fluorescence lifetimes (~10–20 ns), enabling time-gated detection to reject short-lived background fluorescence.
  - LOD: ~0.01 µg/L (10 pg/mL).
- **Capillary electrophoresis with laser-induced fluorescence (CE-LIF):** Ex 405 nm (diode laser).
  - LOD: ~0.05 µg/L.
  - Sample prep: Dilution, filtration.
- **Fluorescence screening device (Lang et al., 2018, JBO):** Custom device with LED excitation at 405 nm for bedside porphyrin quantification in urine.

- LOD: ~5 μg/L.
- No sample prep required.

Raman Detection Porphyrins show characteristic Raman spectra due to the conjugated macrocyclic ring:

Peak (cm^{-1})	Assignment
~1550–1560	Cβ–Cβ stretch (ν_2)
~1490–1500	Cα–Cm stretch (ν_3)
~1370–1380	Pyrrole half-ring (ν_4) — oxidation state marker
~1000–1010	Cα–N stretch
~750–760	Pyrrole breathing mode
~670–680	Porphyrin ring deformation

- **Resonance Raman spectroscopy** (excitation at Soret band ~405 nm) greatly enhances porphyrin-specific signals (10^3–10^4 enhancement).
- **SERS:** Silver or gold nanoparticles with 633/785 nm excitation; LOD in sub-μg/L range reported for metalloporphyrins.
- Conventional Raman is insufficient for trace urinary porphyrins without resonance enhancement.
- **Sample prep:** Acidification, optional extraction.

FTIR Detection Porphyrin infrared bands:

Band (cm^{-1})	Assignment	Notes
~1690–1720	C=O stretch (carboxyl groups)	Multiple COOH groups give strong absorption
~1600–1610	C=C ring stretch	Porphyrin macrocycle
~1400–1450	COO^- symmetric stretch	If deprotonated
~1000–1050	C–N stretch, ring breathing	—
~800–850	C–H out-of-plane bending	—

- ATR-FTIR can detect porphyrins only at elevated concentrations (>100 μg/L). Normal urinary levels are below typical FTIR sensitivity.
- In severely affected porphyria patients (TUP >1000 μg/day), ATR-FTIR might detect porphyrins in dried urine.
- **Research gap:** No validated FTIR method for routine urinary porphyrin quantification.

Voltammetry Detection

- Porphyrins are **electrochemically active** — the conjugated ring system undergoes well-defined oxidation/reduction.
- **Cyclic voltammetry:** Free-base porphyrins show oxidation peaks at **+0.6 to +1.0 V** and reduction at **−1.0 to −1.4 V** vs Ag/AgCl, depending on substituents.
- **Differential pulse voltammetry (DPV):** LOD ~10–50 nM for coproporphyrin on carbon electrodes.
- **Electrochemical HPLC detection** as alternative to fluorescence: amperometric detection at +0.8 V.
- **Sample prep:** Acidification, extraction, or direct injection (HPLC-EC).
- **Limitation:** Selectivity in complex urine matrix is challenging without chromatographic separation.

Other Detection Technologies

1. **Capillary electrophoresis (CE):** UV or fluorescence detection; LOD ~0.05–1 µg/L. Sample prep: dilution, filtration. Fast separation (~10 min).
2. **Lateral flow / paper-based fluorescence:** Experimental devices using UV LED excitation and smartphone cameras to image porphyrin fluorescence on paper strips. LOD ~50–100 µg/L. Low cost, point-of-care potential.
3. **Microfluidic platforms:** Lab-on-chip with integrated fluorescence detection; LOD ~1 µg/L. Miniaturised sample handling.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Spectrophotometry (Soret)	~5–10 µg/L	400–405 nm absorption	Acidification + extraction	Simple, rapid	No fractionation
HPLC-fluorescence (gold standard)	~1 µg/L	Ex 400 / Em 618 nm	Acidification	Full fractionation, isomers	Cost, 30 min
Wood's lamp screening	~100–200 µg/L	Ex 365 / Em 595–630 nm	None	Instant, bedside	Qualitative, insensitive
Direct fluorescence	~0.1 µg/L	Ex 398–410 / Em 595–630 nm	Acidification	Ultra-sensitive	No species ID
Time-resolved fluorescence	~0.01 µg/L	Ex 405 / Em 620 nm	Acidification	Rejects background	Specialised instrument
CE-LIF	~0.05 µg/L	Ex 405 nm laser	Dilution, filtration	Fast, sensitive	CE instrument needed
Resonance Raman	~1 µg/L	Ex 405 nm (Soret resonance)	Acidification	Porphyrin-specific	Specialised, costly
FTIR	>100 µg/L	1690–1720 cm^{-1}	Dried sample	Multianalyte potential	Insensitive for porphyrins
DPV (voltammetry)	~10–50 nM	+0.6 to +1.0 V oxidation	Extraction	Sensitive	Selectivity challenges
Paper-based fluorescence	~50–100 µg/L	UV LED / smartphone	None	Low cost, portable	Semi-quantitative

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